

Cervical lymphadenopathy and neck mass

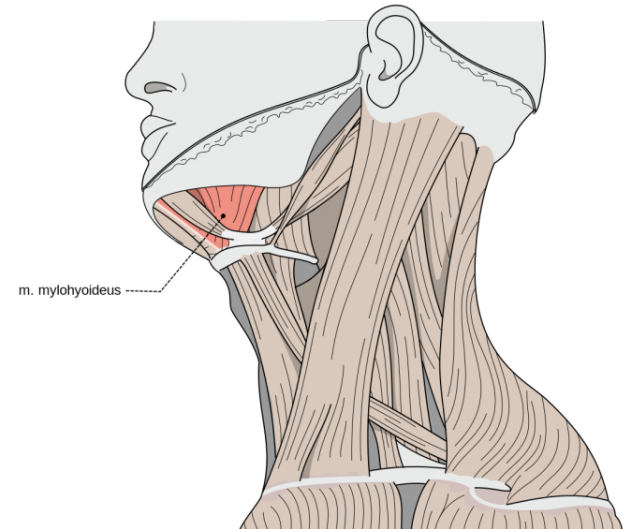
DR:Mariwan latif fatah

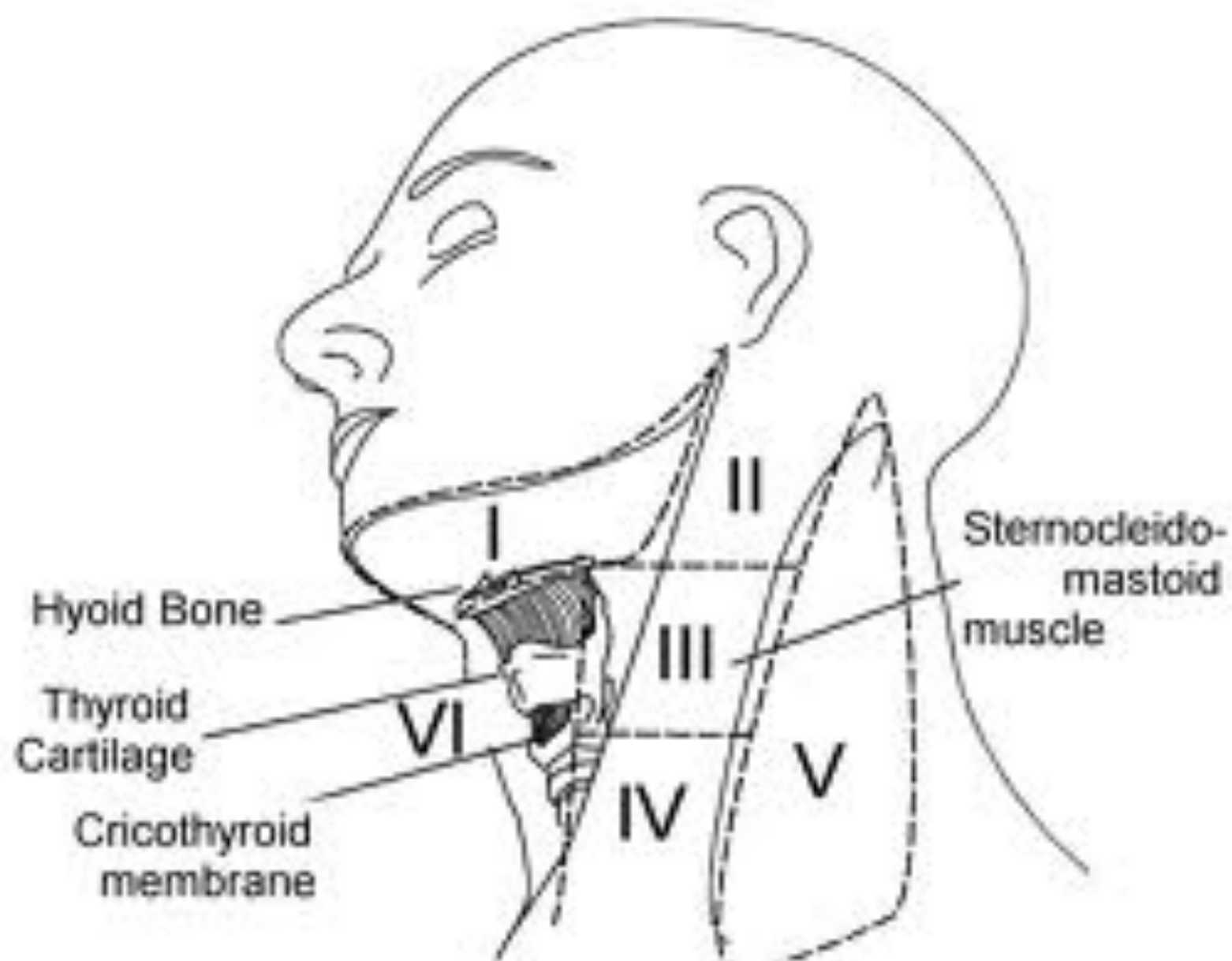
M.B.ch.B F.I.C.M.S

levels of the neck (6 levels)

Level I

- below myohyoid muscle and above the lower margin of the hyoid bone
- anterior to the posterior border of the submandibular glands
 - **level Ia** : sub mental - between the anterior bellies of the digastric muscle
 - **level Ib** : sub mandibular - posterolateral to the anterior belly of digastric
- **drains the lip, oral cavity and tongue**





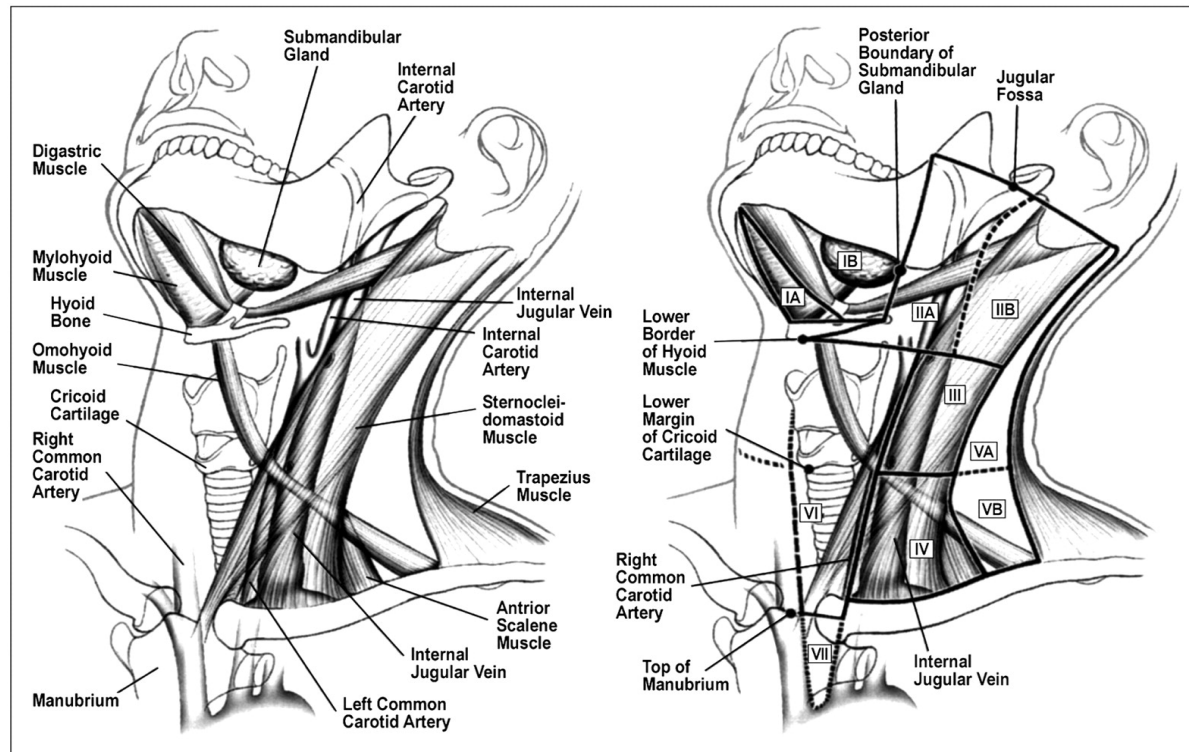
Level II

- jugulodigastric .
- base of skull to lower margin of hyoid bone.
- anterior to the posterior border of sternocleidomastoid (SCM).
- posterior to the posterior border of the submandibular glands .
- **Drains the oropharynx, larynx, hypopharynx and parotid gland.**
- The course of the spinal accessory nerve divides this level into two subzones.
- **Level IIa** lies anteroinferior to the spinal accessory nerve
- **Level IIb** posterosuperior.

Level III

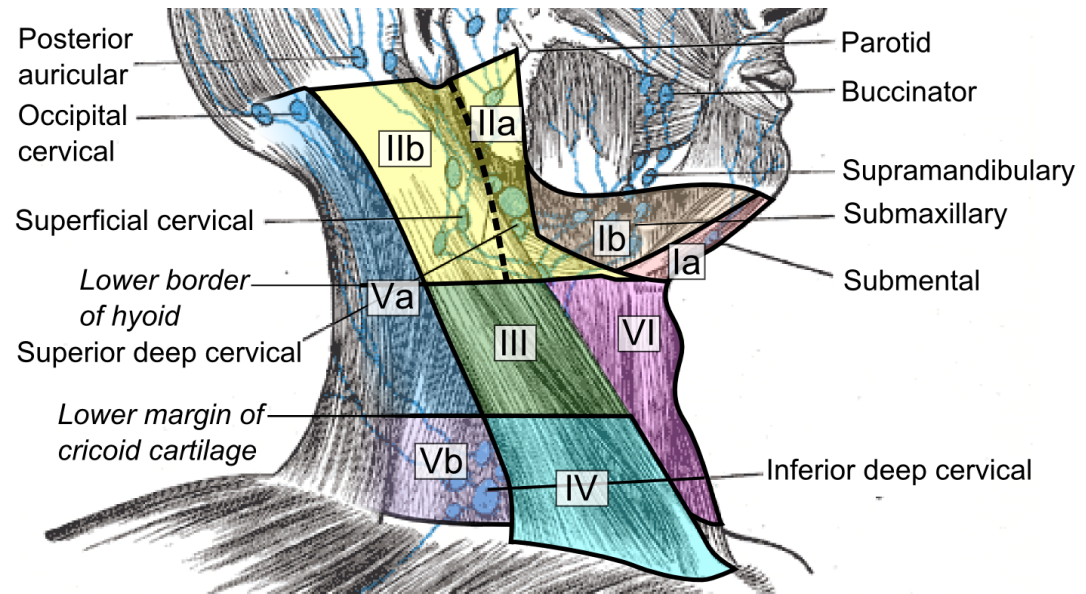
- middle jugular nodes.
- lower margin of hyoid to lower margin of cricoid cartilage.
- anterior to the posterior border of SCM.

✓ **Drains larynx and pharynx.**



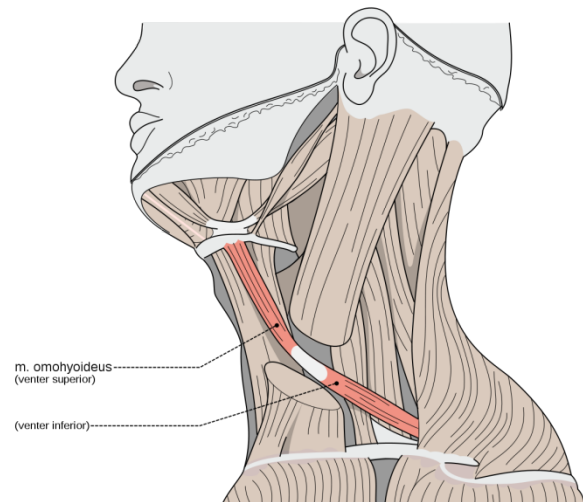
Level IV

- low jugular nodes .
- lower margin of cricoid cartilage to level of the clavicle.
- **Drains larynx and hypopharynx.**



Level V

- posterior triangle nodes .
- It has been divided into sub zones.
- **level Va** lies superior to the inferior belly of the omohyoid muscle .
- **level Vb** level is inferior to the omohyoid muscle.
- The Va area contains the chain of nodes along the accessory nerve, **which drain the nasopharynx.**
- The Vb level contains nodes related to the thyrocervical trunk which **drains the thyroid gland.**

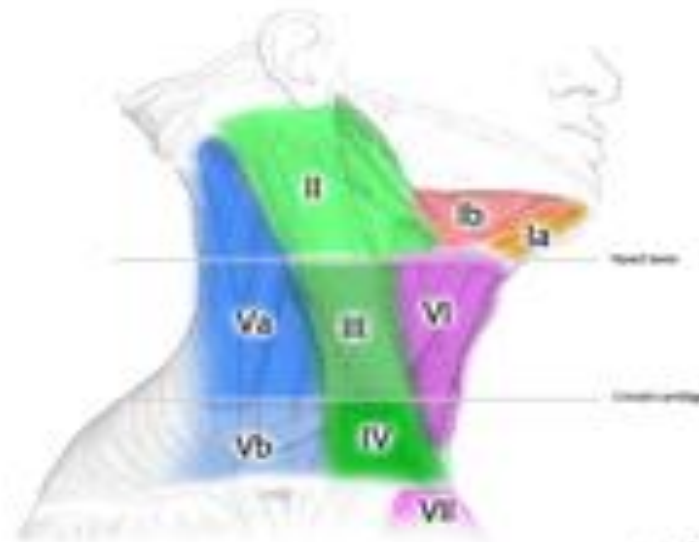


Level VI (anterior or central group)

- prelaryngeal / pretracheal / [Delphian node](#) .
- from inferior margin of hyoid to manubrium.

Level VII

- superior mediastinum.
- between CCAs, below top of manubrium.



Lymph node levels

- Evaluation of a patient with a neck swelling requires a careful and systematic clinical approach with understanding of the anatomical structures of the neck and appreciation of the pathology.

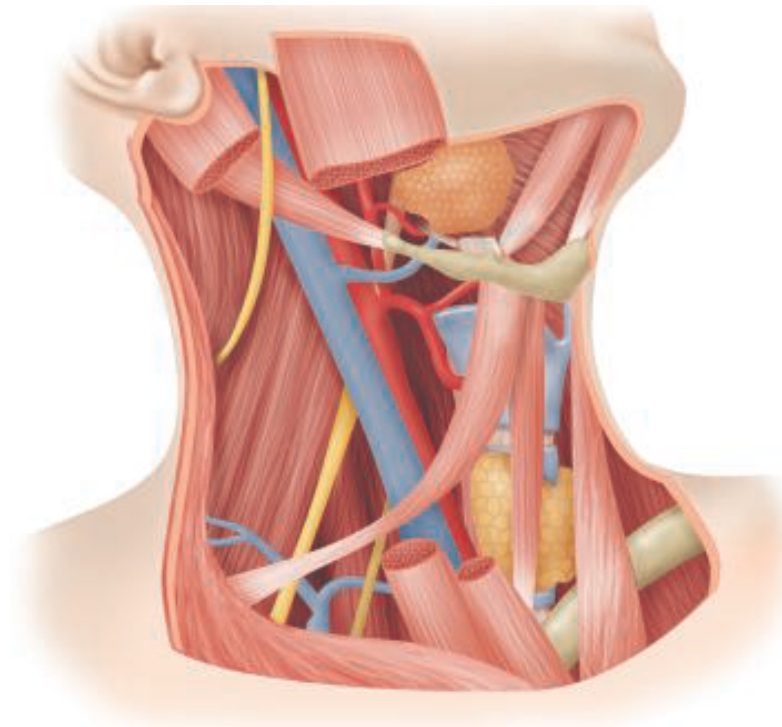
HISTORY

- The key points in the history that will help reach a diagnosis are the **age of the patient, duration and location of the neck mass.**
- Dysphonia, sore throat, dysphagia, odynophagia, referred otalgia and nasal obstruction may suggest a neoplasm of the upper aerodigestive tract.
- Weight loss, anorexia, malaise and night sweats ('B' symptoms) may suggest lymphoma or tuberculosis.

- Past medical history of exposure to tuberculosis, irradiation (risk factor for malignancy),
- occupational/sexual history (e.g. risk factors for human immunodeficiency virus
- foreign travel and contact with pets (e.g. cat scratch disease) are also relevant.

EXAMINATION

- The neck masses should be assessed and recorded in a systematic fashion with emphasis on the number, size, site, shape and consistency.
- It is convenient to use the neck nodal level system to describe the location of lymph node disease in the neck.



- Inspection of the scalp, skin over the face and neck is necessary to look for potential sources of pathology.
- **Otolaryngologists** are best placed to evaluate neck masses as clinical examination is incomplete unless the entire mucosa of the upper aerodigestive tract is examined

- This includes endoscopic assessment of the nasal cavity, nasopharynx, oropharynx, larynx and hypopharynx.
- Where facilities exist, a transnasal oesophagoscopy will allow examination of the oesophagus down to the cardia.
- The examination should conclude with a general physical examination to include the chest, axillae and breasts in women.

- Important signs that should raise concern matted nodes (often felt in lymphoma), nodes that feel hard or rounded (usually indicating malignancy) and ulceration or masses in the upper aerodigestive tract (which can indicate a primary cancer).

INVESTIGATIONS

- In the majority of patients with neck lumps, an ultrasound and a needle biopsy are the first-line investigations.
- If an inflammatory mass is suspected, especially in a young patient, hematological investigations including blood counts, inflammatory markers, viral titers, serology for *Brucella spp.* and *Toxoplasma spp.*

ULTRASOUND IMAGING

- Ultrasound (US) is a useful modality in the evaluation of all lumps, especially of thyroid and salivary Origin.
- with the added ability to precisely target tissue for fine-needle cytology.
- Currently available higher resolution probes can study the architecture of the lump and identify features that have high predictive values for benign or malignant diagnoses.
- It is relatively inexpensive, quick, non-invasive, and does not use ionizing radiation, **but is operator dependent**.



COMPUTERIZED TOMOGRAPHY

- Computerized tomography (CT) is useful in identifying the presence of clinically undetectable lesions (e.g. deep-seated masses in Para pharyngeal and retropharyngeal spaces) and in delineating the extent, vascularity and relationship of neck lumps to adjacent structures.
- It is used routinely for the staging of head and neck tumours and also in the investigation of a patient with an unknown primary presenting with metastatic neck disease.

MAGNETIC RESONANCE IMAGING

- Magnetic resonance imaging (MRI) has uses that are similar to CT, but it also provides very good soft tissue contrast and is the preferred modality where soft tissue detail is important, for instance the **oropharynx and parotid gland**.
- MR angiograms are an excellent modality for vascular delineation

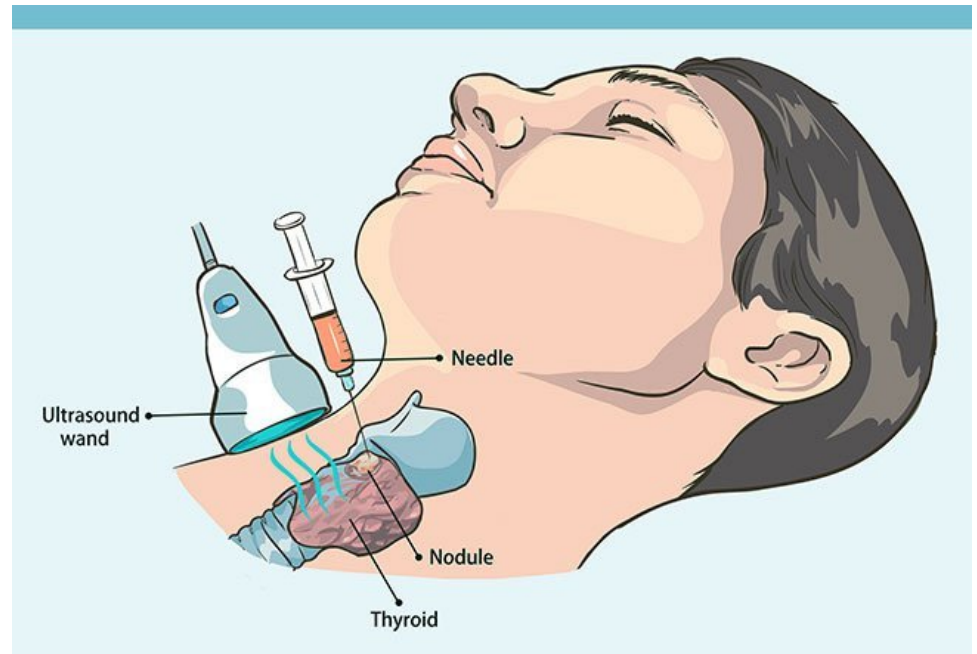


POSITRON EMISSION TOMOGRAPHY

- Positron emission tomography (PET) is a functional imaging technique that depicts tissue metabolic activity and uses short-lived radioisotopes that contain protons that decay emitting positrons.
- PET is rarely used as a sole imaging modality, but PET-CT or PET-MRI is especially useful in assessing patients presenting with metastatic neck disease with an unknown primary and following treatment for head and neck cancer.

FINE-NEEDLE ASPIRATION CYTOLOGY

- Fine-needle aspiration cytology (FNAC) is currently regarded as the key diagnostic modality for neck lumps and can be done with or without US guidance



CORE BIOPSY

- Core biopsy (CB) is a valuable adjunct to the diagnostic process when FNAC is unsuccessful.
- Although there had been earlier fears of seeding the CB needle tract, these are largely unfounded.
- CBs are especially valuable to characterize salivary gland malignancies and lymphomas.



OPEN BIOPSY

- Open biopsy (OB) may be necessary if both FNAC and CB are unhelpful, imaging modalities suggest a malignant process but the primary is unknown or when the diagnosis is a lymphoma.

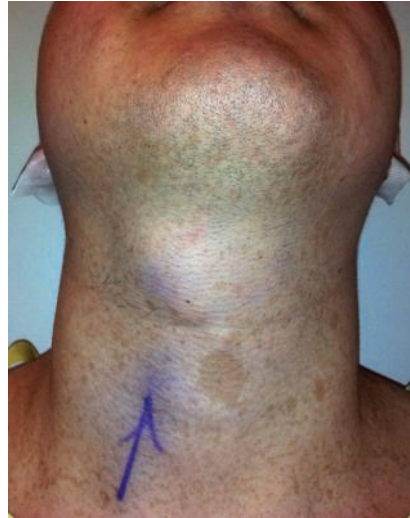
Neck Masses in an Adult

(Midline)

1-Inflammatory thyroiditis.

2-Non-inflammatory

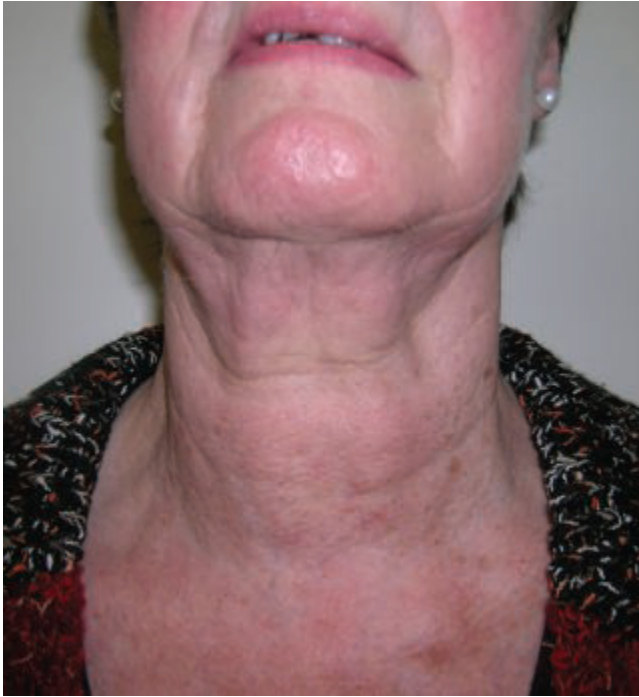
- A. Dermoid cyst.
- B. Thyroglossal duct cyst.
- C. Thyroid nodules.
- D. Thymic cyst.
- E. Lymphoedema.



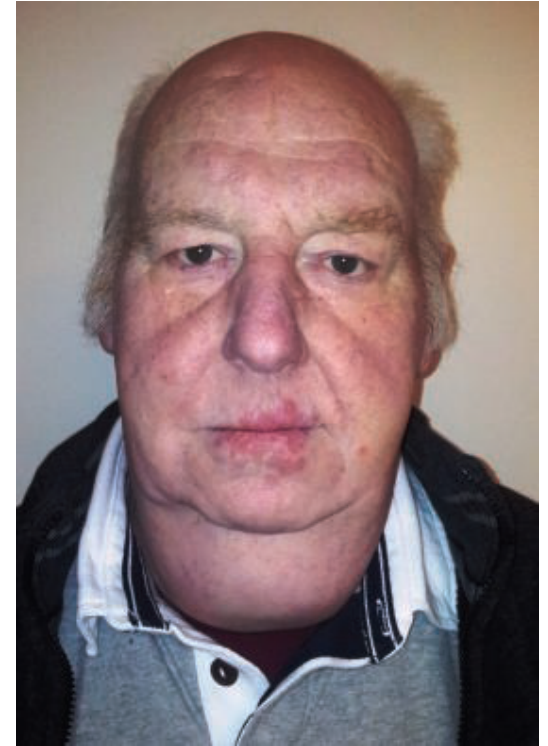
thyroglossal duct cyst

•

Thyroid enlargement



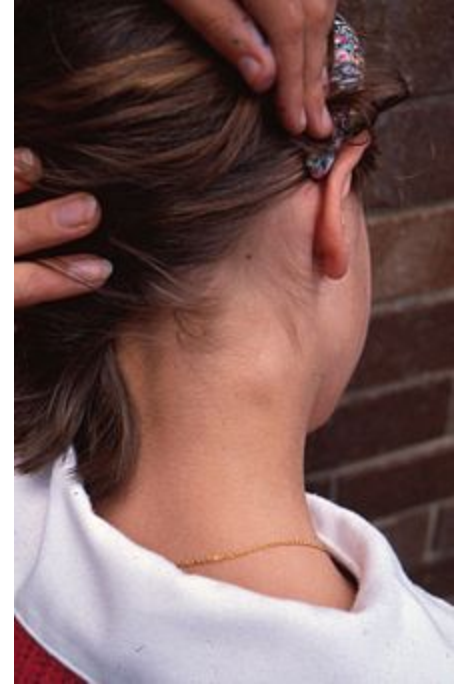
post radiation lymphedema



Lateral neck mass

A. Inflammatory

- Reactive lymphadenopathy
- Infective lymphadenopathy
- (HIV, TB)
- Submandibular sialadenitis
- Parotitis
- Kikuchi's disease
- Kawasaki syndrome



cervical LA

B. Non-inflammatory

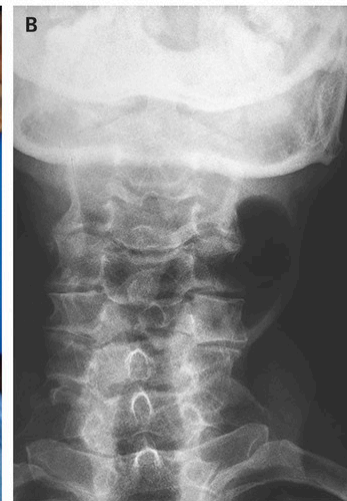
Benign

- Branchial cyst.
- Laryngocoele.
- Parotid tumours.
- Schwannoma.
- Paraganglioma.
- Fibroblastic/myofibroblastic lesions
- Lipoma
- Parapharyngeal space tumours

- Parotid mass



laryngocele



Malignant

- Metastatic squamous cell cancer
- Primary parotid cancers or metastases to lymph nodes
- Metastatic thyroid cancer
- Lymphomas
- Adenocarcinoma from distant sites

APPROACH TO TREATMENT PLANNING

- In the presence of benign pathology, the only treatment available is surgical excision, but patient wishes and expectations are paramount.

- Treatment for head and neck cancer is a rapidly evolving field, and treatment planning is best undertaken in a multidisciplinary setting where appropriate expertise is available.
- national guidelines or local protocols are often used to guide treatment.
- Relying on a positive FNAC alone to plan treatment in patients with head and neck cancer is a strategy beset with hazard, given the reported 3–10 per cent false positive rates.
- **One should always try to ensure that a definitive biopsy is available**

Rule of 7 in the neck

- 7 days—inflammation
- 7 months—neoplasm
- 7 years—congenital defect

Note: The Rule of 7 provides a **probable diagnosis** of the neck mass based on the average duration of the patient's symptoms

*Thank
you*

